INCH-POUND

MIL-C-46228B
23 April 1991
SUPERSEDING
MIL-C-46228A
6 July 1973

MILITARY SPECIFICATION

CARTRIDGE, DELAY, M93 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M93 Delay Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification for
MIL-C-60539	Coating, Anodic, Conventional for Aluminum and Aluminum Alloy

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

5184950	M93 Closed Bomb Assembly

NAVAL ORDNANCE STATION (CAGE Codes 19200 and 19203)

DL8593312	Cartridge, Delay, M93 Assembly
8799620	Box, Packing, Ammunition, For Cartridge, Delay: M93
8799621	Carton, Packing, Ammunition, For Cartridge, Delay: M93
9224944	Carton, Alternative, Packing, Ammunition, For Cartridge, Delay: M70, M93, M113, M155, XM231, OR M293

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. When specified in the contract or purchase order, a first article sample shall be subjected to first article inspection (see 4.4, 6.2, and 6.3).

3.2 Conformance to documents. The M93 Delay Cartridge covered by this specification shall be manufactured in accordance with DL8593312 and all documents listed thereon.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Components.

3.4.1 Primary components. For the purpose of this specification, the M42 Primer, M8 Delay Element, Igniter Charge and the cartridge propellant charge are considered primary components. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

3.4.1.1 Primer. The primer shall conform to the applicable drawing and reference specification. Only primers from lots manufactured within the last 24 months or recertified within the last 12 months shall be used in these units. Prior to assembly into a cartridge, the primer shall be radiographically examined for defects as specified in 4.7.1.

3.4.2 Hardness. The hardness of each component part shall be in accordance with the applicable drawing (see 4.7.2).

3.4.3 Protective finish. The finish of each component part shall be in accordance with MIL-C-60539 and the applicable drawing (see 4.7.3).

3.5 Assembly.

3.5.1 Radiographic examination. Each cartridge assembly shall be radiographically examined for defects (see 4.7.4).

3.5.2 Airtightness. Each assembled cartridge shall exhibit no air leakage greater than 1×10^{-4} standard cubic centimeters per second (see 4.7.5) at a 14.7 psia pressure differential.

3.5.3 Vibration. The assembled cartridge shall withstand vibration as prescribed in MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I 3.2.5 without evidence of any disintegration of the primer, propellant, or igniter, or derangement of any component (see 4.7.7 and 6.2).

3.6 Functional requirements.

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3.6.1 Primer sensitivity. The primer shall fire when drop tested at an energy level of 26 inch-ounces in a sectioned cartridge void of igniter and propellant charge (see 4.7.8.1).

3.6.2 Cartridge performance.

3.6.2.1 Misfires. The cartridge shall not fail to fire.

3.6.2.2 Ballistics. The cartridge, when tested in a closed bomb (Drawing 5184950), shall produce the pressures, delay times (see 6.6.1), and times to P_{max} (see 6.6.2) at temperatures specified in table I (see 4.7.8.2).

TABLE I. Ballistic requirements.

Temperature (•F)	Delay time (sec)		Pressure (psig)		Time to P_{max} (ms)
	Lower	Upper	Lower	Upper	
- 60 to - 70	.85	1.30	2300	3400	50
65 to 75	.85	1.20	2300	3400	50
195 to 205	.80	1.15	2300	3400	50

3.7 Workmanship. Cartridges shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified in the contract or purchase order (see 6.2), inspection conditions shall be as follows:

a. Temperature: Room ambient 20• to 25• C (68• to 77• F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric: 14.7 psia

4.4 First article inspection. At the beginning of regular production, the contractor shall submit a first article sample to a Government approved facility for evaluation. The sample shall consist of 21 assembled cartridges for ballistics testing including 12 for vibration tests, 5 assembled cartridges for sealant tests, 10 sets of unassembled components for dimensional and visual examination, and the quantity specified in accordance with 4.4.2, i.e., cartridges assembled without the igniter and propellant charge for primer sensitivity test. The first article sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. When it becomes necessary to make a change in either the primer, propellant, or igniter lot after initial production, 12 assembled cartridges from the first production lot thereafter shall be tested in accordance with 4.7.7. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).

4.4.1 Examination, tests, and facility. After provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract (see 6.2).

4.4.2 Sample size for primer sensitivity. The number of cartridges assembled without igniter or propellant to be submitted for primer sensitivity tests shall be in accordance with table II.

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TABLE II. Primer sensitivity.

Lot size	Sample size
3 to 15	3
16 to 25	4
26 to 40	5
41 to 65	7
66 to 110	10
111 to 180	15
181 to 300	25
301 to 500	30
501 to 800	35
801 to 1300	40
Over 1300	50

4.4.3 Temperature conditioning for ballistic test. The sample shall be temperature conditioned and tested to determine compliance with 3.6.2.2. The sample for firing shall be conditioned to table III.

TABLE III. Temperature conditioning.

Temperature (•F)	Not vibrated	Vibrated
-60 to -70	3	4
65 to 75	3	4
195 to 205	3	4

4.4.4 First article failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in first article sample disapproval.

4.5 Quality conformance inspection.

4.5.1 Lot.

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4.5.1.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105.

4.5.1.2 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not more than 1200.

4.5.2 Examination and tests. The quality conformance sample shall be subjected to the examinations of 4.6 and all the test procedures of 4.7 except for 4.7.7.

4.6 Examination. All cartridges and components shall be examined visually and radiographically and characteristics defined in accordance with the applicable drawings. One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter. When compliance is in doubt as a result of visual inspection, the characteristic shall be measured or gaged to determine acceptability. All nonconforming items shall be rejected.

4.6.1 Cartridge, Delay, M93 Assembly, Drawing 8593312.

Characteristic	Defect	Method of Inspection
Critical		
1	Any part missing	X-ray/Visual
2	Improper assembly	X-ray/Visual
3	Mixed delay elements or propellant	Visual
4	Metal defective	Visual
5	Crimp missing or incorrect	Visual
6	Marking missing, incorrect, or illegible	Visual
Major		
101	Cartridge profile failure	Gage
102	Diameter of body, max	Gage
103	Diameter of head, max	Gage
104	Location of datum diameter	Gage
105	"O" ring split, cracked or cut	Visual
106	Protective finish inadequate or defective	Visual
107	Foreign matter present	Visual
108	Burr	Visual
Minor		
201	Total length, max	Gage
202	Case damaged	Visual
203	Varnish inadequate or missing	Visual

4.6.2 Delay Element, M8, Drawing 8593163, covering a detail of Drawing 8593312.

Characteristic	Defect	Method of Inspection

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Critical		
1	Mixed delay elements	Visual
2	Flash hole missing or obstructed	Visual
3	Primer uncrimped or seal missing	Visual
4	Marking missing, incorrect, or illegible	Visual
5	Delay charge not visible through flash hole	Visual
6	Metal defective	Visual
Major		
101	Protective finish inadequate or defective	Visual
102	Burr	Visual
103	Foreign matter present	Visual
Minor		
None defined		

4.6.3 Case, Cartridge, Drawing 8595202, covering a detail of Drawing 8593312.

Critical		
1	Metal defective	Visual
Major		
101	Concentricity of head with body	Gage
102	Straightness of base	Gage
103	Diameter of body, min	Gage
104	Diameter of head, min	Gage
105	Thickness of base flat	Gage
106	Thickness of wall	Gage
107	Protective finish inadequate or defective	Visual
108	Burr or sharp corner	Visual
109	Foreign matter present	Visual
Minor		
201	Diameter of head, max	Gage
202	Hardness	Gage
203	Total length	Gage
204	Radius or chamfer missing or incorrect	Visual
205	Surface finish improper	Visual

4.6.4 Carton, Packing, Ammunition, Drawing 8799621.

Characteristic	Defect	Method of Inspection

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Critical
None defined

Major
None defined

Minor

201	Contents loose		Manual/Visual
202	Separator or filler missing		Visual
203	Heat sealer incomplete, loose, or distorted	Visual	
204	Cut or damaged through impregnated box		Visual
205	Marking missing, incorrect, or illegible		Visual

4.6.5 Box, Packing, Ammunition, Drawing 8799620.

Critical
None defined

Major
None defined

Minor

201	Contents loose		Manual/Visual
202	Strapping missing, loose, broken, or improperly assembled		Visual
203	Contents of box exposed		Visual
204	Box damaged		Visual
205	Marking missing, incorrect, or illegible	Visual	
206	Number of cartons incorrect		Visual

4.6.6 Carton, Alternative, Packing, Drawing 9224944.

Critical
None defined

Major
None defined

Minor

201	Contents loose		Manual
202	Box broken or damaged	Visual	
203	Number of units contained incorrect		Visual

Characteristic	Defect		Method of Inspection
204	Marking missing, incorrect, or illegible	Visual	

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205	Tape incomplete	Visual

4.7 Test methods and procedures.

4.7.1 Primer x-ray. Prior to assembly into the cartridge, the primer shall be radiographically examined in accordance with MIL-STD-453. Each primer shall be x-rayed on boards or trays with the cup side perpendicular to the rays. The x-ray negatives shall be identified by item nomenclature and by lot number. Primers are to be left mounted to the boards or trays until after the x-rays have been reviewed and defects identified to facilitate removal of defective units. After screening, the x-ray negatives with defectives identified, shall be forwarded to the engineering activity designated by the acquiring activity. Interpretation of the radiographic films shall verify the following:

Presence of Primer Charge
No Material Defects

4.7.2 Hardness. Ten units from each heat-treated lot of components shall be tested with a hardness tester. Failure of a unit to comply with the requirements of the drawings shall be cause for rejection of the lot of components.

4.7.3 Protective finish. Unless otherwise specified (see 6.2 and 6.4), sampling plans and procedures in the determination of acceptability of coated parts shall be in accordance with the provisions set forth in MIL-C-60539.

4.7.4 Radiographic examination. All assembled cartridges shall be radiographically examined in accordance with MIL-STD-453. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph plate identification shall include item nomenclature, the complete lot number as stamped on the cartridges, the span of serial numbers displayed, and the contract number. Cartridges having any observable imperfections in assembly shall be cause for rejection. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Discontinuities in serial numbers shall be identified on the x-ray negative. Defective cartridges, found by the contractor, shall be clearly identified on the negative and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridge with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings (see 6.2).

4.7.5 Airtightness. Each cartridge shall be subjected to specified pressure differential. The method for determining airtightness shall be with a helium mass spectrometer. The helium leak rate shall be converted to standard cubic centimeters of air. Failure to comply with the requirements of 3.5.2 shall be cause for rejection of the cartridge.

4.7.6 Sealant application. Five cartridges for first article inspection shall be disassembled in such a manner that determination can be made that the sealant has been properly applied for the full length of the

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joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

4.7.7 Vibration test. Twelve of the 21 assembled cartridges of the first article sample, x-rayed as directed in 4.7.4, properly identified, shall be subjected to vibration. The method and procedures for performing the vibration test shall be in accordance with MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I-3.2.5 (see 6.2). Then the 12 vibrated cartridges shall be x-rayed again. Evidence of any disintegration of the primer, propellant, or igniter, or derangement of any component shall be cause for rejection of the sample.

4.7.8 Functional. The following tests shall be performed by the Government.

4.7.8.1 Primer sensitivity. A quantity of cartridges, in accordance with table II, void of delay charge igniter and propellant, from the first article sample and each subsequent lot shall be tested for sensitivity. In the event loaded cartridges are selected for test, remove and discard all powder charges from the test cartridge by sectioning the cartridge case around the periphery, ½ inch maximum above the base. Drop test at 13 inches (adjusted) using a steel ball 15/16-inch in diameter, weighing 1.94 ± 0.02 ounces, in a primer sensitivity machine with a cartridge holder approved by the acquiring agency. The height of 13 inches shall be adjusted taking into consideration the actual weight of the ball so that the energy level specified in 3.6.1 is achieved. Failure of the sample to comply with the requirements of 3.6.1 shall be cause for rejection of the sample or the lot.

4.7.8.2 Ballistics.

4.7.8.2.1 First article. Twenty-one cartridges of the first article sample shall be grouped and temperature conditioned as described in table III and 4.8. Following removal from the conditioning chamber, the cartridge shall be tested as required by 3.6.2.2. Failure of a cartridge to comply with the requirements of 3.6.2.2 shall be cause for sample rejection.

4.7.8.2.2 Production lot acceptance. Twenty-one assembled cartridges from each production lot shall be divided into three equal groups, temperature conditioned, and fired in accordance with 3.6.2.2. If the sample is not evenly divisible by three, the remaining cartridges will be tested at -60• and 205• F. Failure of a cartridge to comply with 3.6.2.2 shall be cause for rejection of the lot. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each production lot for investigative purposes.

4.7.8.3 Retest. There shall be no retests.

4.7.8.3.1 Test failure. If a test failure is attributable to an assignable cause, excluding the test cartridge, the original test results shall be discarded and that test reconducted.

4.8 Temperature conditioning. The conditioning chamber shall be preconditioned to the specified temperature. Once the temperature inside the chamber has stabilized, the samples shall be placed inside and maintained at the specified temperature for 4 hours minimum, 24 hours maximum.

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4.8.1 Reconditioning. The sample must be tested within five minutes after removal from the conditioning chamber. If this time is exceeded, the cartridge shall be reconditioned in accordance with 4.8.

4.9 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8799620 and 8799621. Level B shall be in accordance with Drawings 8799621 and 9224944.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The assembly is used in cartridge actuated devices for Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date, and revision letter of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced and that MIL-STD-810D shall be used (see 2.1, 3.5.3, and 4.7.7)

c. Whether a first article sample is required (see 3.1, 4.4, and 6.3)

d. Test facility for first article tests (see 4.4.1)

e. Production test activity and lot size (see 4.1 and 4.5.1.2)

f. Any special test conditions (see 4.3)

g. Directions for sampling for classification of defects (see 4.6)

h. Directions for shipping radiographs of primers and of entire lot of cartridge assemblies (see 4.7.1 and 4.7.4)

i. Directions for sampling for protective finish tests (see 4.7.3)

j. Level of preservation and packing (see 5.1)

k. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," (DOD 4145.26M) are applicable. NOTE: When

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this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition and Explosives Ashore," (OP 5) are applicable.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Explosive hazard classification information.

Hazard Classification Division: 1.4
Storage Compatibility Group: S
DOT Class: C
DOT Marking - Explosive Power Device - Class C - Handle Carefully - Keep Fire Away

6.5 Hazard notice. The cartridge described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.6 Definitions.

6.6.1 Delay time. Delay time is defined as the time from primer impact to the start of pressure rise.

6.6.2 Time to P_{max}. Time to P_{max} is defined as the time that elapses between the first indication of pressure to maximum pressure as recorded on a pressure time record.

6.7 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.8 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

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Custodian:
Navy-OS
Air Force-99

Preparing activity:
Navy-OS
(Project 1377-0984)

Review activity:
Air Force-70

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